

Vladimir LAPIN, Ph.D.

Senior Research Engineer | Climate Scientist | Applied Mathematician

PERSONAL DATA

NATIONALITY: Russian | Authorized to work in the EU (Permanent Resident – Spain)
LOCATION: Barcelona Supercomputing Center, Barcelona, Spain
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PROFILES: [Website](#) | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#) | [ORCID](#)

PROFILE SUMMARY

- **Physics–ML Integration Specialist:** Applied Mathematician (1st Honours, MSU) with 15 years in climate model development, specializing in coupling neural network emulators with physical dynamical cores.
- **AI Adoption & Verification:** Leading AI adoption across the research group; architecting automated verification harnesses and workflows transforming how scientific software is maintained.
- **Production Infrastructure:** Co-lead for Auto-EC-Earth4 HPC ensemble workflows (Autosubmit/Slurm) – the scalable execution pipeline required for hybrid physical–ML earth system models.

EMPLOYMENT

- | | |
|-------------|---|
| 2025 – now | Senior Research Engineer, Earth Sci. Dept., BSC, Spain
ROLE <ul style="list-style-type: none">◦ Production Workflows & AI Adoption: Co-leading Auto-EC-Earth4 ensemble workflows on HPC clusters (Autosubmit/Slurm); leading AI adoption across the research group.◦ Consortium Maintainer: Co-maintaining the EC-Earth4 model repository across 30 institutes in 12 European countries; integration standards, CI, and release coordination.◦ ML Integration: Advising on emulator adoption for EC-Earth4 – identifying parametrizations where learned emulators are viable, defining coupling interfaces, and contributing domain knowledge to a CNN-based biogeochemistry emulator project. |
| 2019 – 2024 | Recognized Research Engineer, Earth Sci. Dept., BSC, Spain
ROLE <ul style="list-style-type: none">◦ Model Development & Data Assimilation: Development of EC-Earth3; designed nudging scheme for sea-ice state estimation and maintained assimilation pipelines for ocean and atmosphere components.◦ Ensemble Infrastructure: Built high-throughput ensemble systems (50–100 SYPD, 10s–100s members) on HPC clusters – generating the large simulation datasets that ML emulator training depends on. |
| 2015 – 2017 | Postdoctoral Scientist, Ocean Dept., MPI-M, Germany
ROLE <ul style="list-style-type: none">◦ Sea-ice Modelling: Developed the sea-ice component of the ICON climate model, focusing on numerical stability, algorithm efficiency and physical realism. |
| 2011 – 2014 | Research Fellow, Applied Math Dept., Univ. of Leeds, UK
ROLE <ul style="list-style-type: none">◦ Numerical Modelling: Developed high-resolution global numerical models for ocean tides (shallow water equations), implementing an improved finite difference scheme.◦ Parallel Computing: Investigated multi-scale internal tide generation over rough topography using parallel computing (domain decomposition).◦ Data Analysis: Performed extensive data analysis and visualization of global model outputs in MATLAB.◦ Teaching: Teaching Assistant for Calculus of Variations, Calculus & Mathematical Analysis, and Modelling with Differential Equations. |

SUM 2008	Risk Analyst (Intern), Raiffeisenbank, Russia
ROLE	◦ Data Analysis: Portfolio risk diagnostics, quantitative metric tracking, and preparation of operational risk reports.
SUM 2007	Junior Statistician, ACNielsen, Russia
ROLE	◦ Statistics: Statistical analysis of retail consumer purchase data and anomaly detection in store audit datasets.

SKILLS

MODELS	Ocean/climate/fluid dynamics, numerical model development, PDEs, data assimilation
CODE	Python, C/C++, FORTRAN (MPI, OPENMP), GIT, YAML, JSON, MATLAB, COMSOL
DATA	Python (xarray, numpy, pandas), NETCDF, CDO, High-performance Data Pipelines
HPC / AI	Linux/BASH, SLURM, Workflow Engines (AUTOSUBMIT), Agentic Workflows (GH-AW, agent contracts), LMOD/CONDA, CI/CD
ML / AI	AI agent workflows (GH-AW, agent contracts), PyTorch, Physics-Informed Neural Networks (PINNs), emulator design & coupling, hybrid model drift validation
CONSORTIA	Co-maintainer of EC-Earth4 across 30 partner institutes; operational release coordination, integration standards, CI/CD, documentation
MENTORING	Supervision of 2 PhD and 3 Master's students; organization of training events
OUTPUT	27 peer-reviewed publications; 21 international presentations

EDUCATION

2008 – 2011	Ph.D. in Applied Mathematics, University of Limerick, Ireland
ROLE	◦ Applied Math: Research in geophysical fluid dynamics and thin-film Stokes flows using asymptotic analysis and numerical modelling (MATLAB, COMSOL). ◦ Teaching: Teaching Assistant for Engineering Mathematics I/II, Vector Analysis, and Mathematical Modeling and Tutor in the Maths Learning Center. ◦ Thesis: “Resonant over-reflection of waves by jets in a rotating ocean.”
2003 – 2008	B.Sc., M.Sc. in Applied Mathematics, Moscow State University, Russia
	◦ Grades: 4.85/5*, 1st Honours ◦ Thesis: “On the stability of the plane flows of visco-plastic fluids”

SELECTED PUBLICATIONS

2024	Silvy, Y et al. (Dec. 2024). “AERA-MIP: emission pathways, remaining budgets, and carbon cycle dynamics compatible with 1.5 and 2 °C global warming stabilization”. <i>Earth System Dynamics</i> 15 , 1591–1628. ISSN: 2190-4987.
2022	Hermanson, L et al. (Apr. 2022). “WMO Global Annual to Decadal Climate Update: A Prediction for 2021–25”. <i>Bulletin of the American Meteorological Society</i> 103 , E1117–E1129. ISSN: 1520-0477.
2020	Tsujino, H et al. (Aug. 2020). “Evaluation of global ocean–sea-ice model simulations based on the experimental protocols of the Ocean Model Intercomparison Project phase 2 (OMIP-2)”. <i>Geoscientific Model Development</i> 13 , 3643–3708. ISSN: 1991-9603.
2014	Stammer, D et al. (Aug. 2014). “Accuracy assessment of global barotropic ocean tide models”. <i>Reviews of Geophysics</i> 52 , 243–282. ISSN: 8755-1209.
2011	Lapin, VN (Feb. 2011b). “Stability of the Couette flow of ideal rigid-plastic bodies”. <i>Moscow University Mechanics Bulletin</i> 66 , 1–7. ISSN: 1934-8452.

REFERENCES

Group leader: [Dr. Pablo Ortega Montilla](#)
Dept. of Earth Sciences
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Postdoc Advisor: [Dr. Stephen Griffiths](#)
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Group leader: [Dr. Johann Jungclaus](#)
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Max Planck Institute for Meteorology (MPI-M)
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Ph.D. Supervisor: [Prof. Eugene Benilov](#)
Dept. of Mathematics & Statistics
University of Limerick
eugene.benilov@ul.ie

EXTRA

PUBLICATIONS

IN PREP	Lapin, VN & Griffiths, SD (n.d.). "Improved accuracy of finite difference ocean models with irregular boundaries." <i>Ocean Model.</i> ()
2026	Galí, M et al. (2026). "Convection injects labile particulate organic carbon to the deep ocean". <i>Science Advances</i> Orihuela-García, MA et al. (2026). "Dissecting mesopelagic particulate organic carbon budgets in the North Atlantic: A mechanistic diagnosis and evaluation of PISCESv2_RC". <i>Bio-geosciences</i> Saurral, RI et al. (2026). "The key role of Mediterranean and North Atlantic sea surface temperatures on the 2024 record-breaking Valencia precipitation event". <i>Weather and Climate Extremes</i>
2025	Li, H et al. (2026). "Multi-model reconstructions and predictions of the CO2 fluxes and atmospheric CO2 variations". <i>Bulletin of the American Meteorological Society</i> Carréric, A et al. (Oct. 2025). "Comparing the seasonal predictability of the Tropical Pacific variability in EC-Earth3 at two horizontal resolutions". Bilbao, R et al. (2025). "The sensitivity of EC-Earth3 decadal predictions to the choice of volcanic forcing dataset: insights for the next major eruption". <i>Geoscientific Model Development</i> Mahmood, R et al. (Mar. 2025). "Multi-decadal initialized climate predictions using the EC-Earth3 global climate model". Cvijanovic, I et al. (Mar. 2025). "Arctic sea-ice loss drives a strong regional atmospheric response over the North Pacific and North Atlantic on decadal scales". <i>Communications Earth & Environment</i> 6 , ISSN: 2662-4435.
2024	Carreric, A, Ortega, P, Lapin, V et al. (Jan. 2025). "Comparing the seasonal predictability of Tropical Pacific variability in EC-Earth3 at two different horizontal resolutions". Silvy, Y et al. (Dec. 2024). "AERA-MIP: emission pathways, remaining budgets, and carbon cycle dynamics compatible with 1.5 and 2 °C global warming stabilization". <i>Earth System Dynamics</i> 15 , 1591-1628. ISSN: 2190-4987. Bernardello, R et al. (2024). "Ocean biogeochemical reconstructions to estimate historical ocean CO2 uptake". <i>Earth System Dynamics</i> Carreric, A, Ortega, P, Doblas-Reyes, F et al. (2024). "Comparing the seasonal predictability of ENSO and the Tropical Pacific variability in EC-Earth3 at two different horizontal resolutions"
2022	Simon, A et al. (Aug. 2022). "Pacific Decadal Oscillation modulates the Arctic sea-ice loss influence on the midlatitude atmospheric circulation in winter". <i>Weather and Climate Dynamics</i> 3 , 845-861. ISSN: 2698-4016. Acosta Navarro, JC et al. (May 2022). "Added value of assimilating springtime Arctic sea ice concentration in summer-fall climate predictions". <i>Environmental Research Letters</i> 17 , 064008. ISSN: 1748-9326. Hermanson, L et al. (Apr. 2022). "WMO Global Annual to Decadal Climate Update: A Prediction for 2021-25". <i>Bulletin of the American Meteorological Society</i> 103 , E1117-E1129. ISSN: 1520-0477.
2020	Tourigny, E et al. (Mar. 2022). "Near-term prediction of the global carbon cycle using EC-Earth3-CC, the Carbon Cycle version of the EC-Earth3 Earth System Model". Tsujino, H et al. (Aug. 2020). "Evaluation of global ocean-sea-ice model simulations based on the experimental protocols of the Ocean Model Intercomparison Project phase 2 (OMIP-2)". <i>Geoscientific Model Development</i> 13 , 3643-3708. ISSN: 1991-9603.
2014	Stammer, D et al. (Aug. 2014). "Accuracy assessment of global barotropic ocean tide models". <i>Reviews of Geophysics</i> 52 , 243-282. ISSN: 8755-1209. Benilov, ES & Lapin, VN (Dec. 2014). "An example where lubrication theory comes short: hydraulic jumps in a flow down an inclined plate". <i>Journal of Fluid Mechanics</i> 764 , 277-295. ISSN: 1469-7645.

- 2013 | Benilov, ES & Lapin, VN (Nov. 2013). "Inertial instability of flows on the inside or outside of a rotating horizontal cylinder". *Journal of Fluid Mechanics* **736**, 107–129. ISSN: 1469-7645.
- Benilov, E & Lapin, V (2013). "On resonant over-reflection of waves by jets". *Geophysical and Astrophysical Fluid Dynamics*
- 2011 | Lapin, VN (2011a). "Resonant over-reflection of waves by jets". PhD thesis. URL: <http://hdl.handle.net/10344/1965>
- Benilov, ES, Lapin, VN & O'Brien, SBG (Nov. 2011). "On rimming flows with shocks". *Journal of Engineering Mathematics* **75**, 49–62. ISSN: 1573-2703.
- Benilov, E & Lapin, V (2011). "Shock waves in Stokes flows down an inclined plate". *Physical Review E - Statistical, Nonlinear, and Soft Matter Physics*
- Lapin, VN (Feb. 2011b). "Stability of the Couette flow of ideal rigid-plastic bodies". *Moscow University Mechanics Bulletin* **66**, 1–7. ISSN: 1934-8452.
- 2009 | Ward, JA, Lapin, VN & Lee, W (2009). "The Effect of Mechanical Loading on the Frequency of an Oscillator Circuit." In: *Proc. 70th Eur. Study Gr. with Ind.*

B.Sc., M.Sc. in APPLIED MATHEMATICS and MECHANICS, 1st Honours

M.Sc. at the Department of **Mechanics of Composites**

COURSE	HOURS PER TERM	GRADE*
Mathematical Analysis I, II, III, IV	144/128/144/96	5/4/5/5
Complex Analysis I, II	72/64	5/5
Functional Analysis I, II	72/64	5/5
Abstract Algebra	108	5
Discrete Mathematics I, II	72/64	5/5
Analytic Geometry	144	5
Linear Algebra and Geometry	128	5
Differential Geometry and Topology I, II	72/64	5/5
Ordinary Differential Equations I, II	72/64	P/5
Partial Differential Equations I, II	72/64	P/4
Numerical Methods I, II	90/64	P/5
Theory of Probability and Mathematical Statistics	72	5
Theory of Random Processes	64	5
General Physics I, II	48/72	P/5
Theoretical Mechanics I, II, III	90/96/72	5/P/4
Mechanics of Continua I, II, III	96/90/80	5/5/5
Optimal Control of Mechanical Systems I, II	54/48	P/4
Statistical Physics	36	4
Laboratory Practice [‡] I, II, III, III	54/48/54/48	P/5
Programming (C/C++) I-VI	54/80/90/32/36/32	P/P/5/P/P/P
History of Mathematics and Mechanics	54	P
English Language	72/64/72/64	P/P/P/5
National History I, II	54/54	P/5
Civil Defence	32	P
Economics I, II	64/54	P/5
Sociology	54	P
Philosophy I, II	54/48	P/4
Physical Training (4 years)	476	P
Modeling and distance control of robots [†]	36	5
Neural networks [†]	36	5
Geometry in Quantum Mechanics [†]	36	5
Applied Optimal Control [†]	32	5
Averaging Methods [†]	36	5
Tensor analysis [†]	36	5
Advanced Solid Mechanics [†]	64	5
Composite Mechanics [†]	36	5
Theory of viscoelasticity [†]	32	5
Stability Problems in Mechanics of Continua [†]	32	5
Degree Thesis		5
Total Hours and GPA	4866	4.85/5**

* Russian universities implement a [five point grading system](#), where “5” is “Excellent” and “2” is “Fail”. Some courses are graded on “Pass/Not pass” basis and here “P” stands for “Pass”.

** Equivalent US GPA is 3.88 (when crudely converted to GPA 4.0 scale).

† Indicates an optional course; other courses are part of the compulsory curriculum for the degree.

‡ In the laboratories of the [Institute of Mechanics, MSU](#).